XIAMEN UNIVERSITY MALAYSIA

ONLINE FINAL EXAMINATION

Course Code:	COM106			
Course Name:	Electrical Engineering			
Question Paper Setter:	Assoc. Prof. Ir. Ts. Dr. Wong Shen Yuong			
Academic Session:	2021/09	Question Paper:	A ☑	B ☐
Total No. of Pages:	5	Time Allocated:	2 hours	
Additional Materials:	-			
Apparatus Allowed:	Calculator			

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 5 Questions.

   Answer **ALL** questions.
2. Your answers must be handwritten.
3. Please follow the requirement of each question and paste on the downloadable answer book provided.
4. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

NOTE: ALL ANSWERS IN THE FORM OF DECIMAL (0.000), NOT fraction ($\frac{X}{Y}$).

Question 1 [20 marks]

(CLO 1, PLO 1, C3)

Referring to Figure 1,

(a) Solve equivalent resistance, Req. [5 marks]

(b) Solve V_1, V_2, V_3. [9 marks]

(c) Solve I_1, and I_4. [6 marks]

Please show all necessary diagrams.

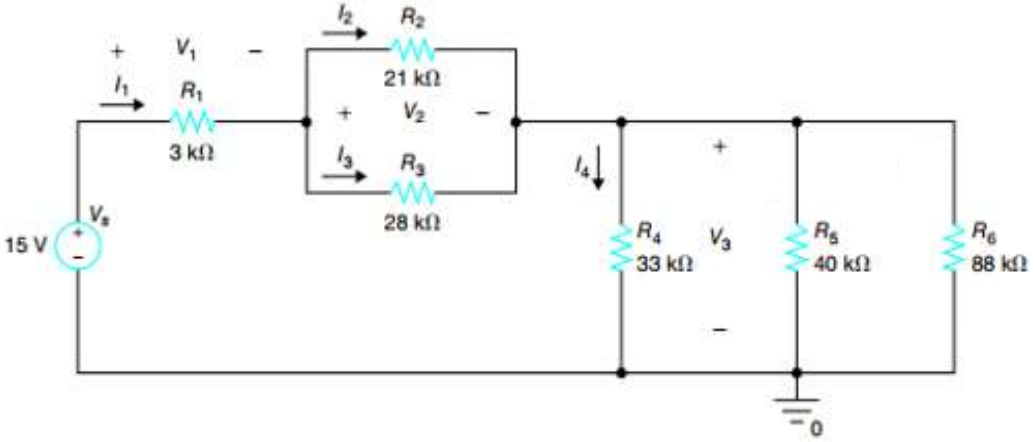


Figure 1

Question 2 [14 marks]

(CLO 1, PLO 1, C3)

Referring to Figure 2, apply mesh analysis to find currents i_1, i_2, i_3 in the circuit.. [14 marks]

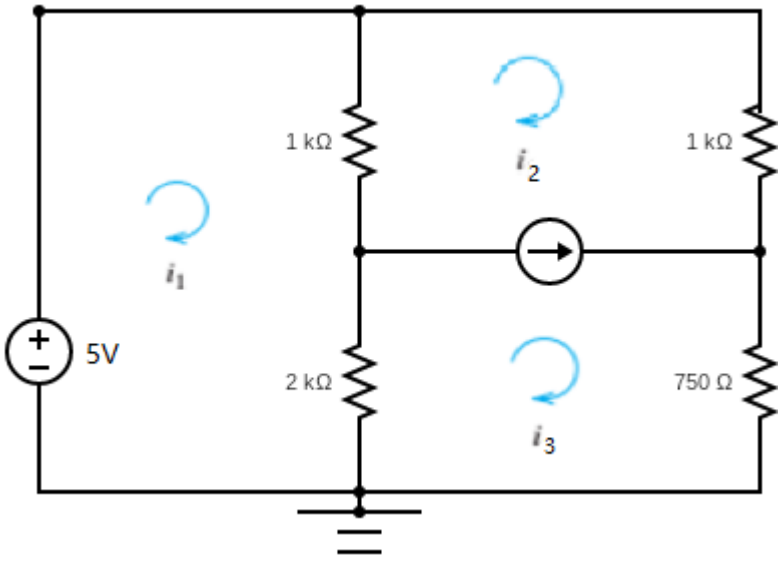


Figure 2

Question 3 [24 marks]

(CLO 2, PLO 2, C4)

Referring to Figure 3,

(a) Find the Thevenin and Norton equivalent. [18 marks]

(b) Refer to you answer in (a), calculate the maximum power that can be delivered to the load. [4 marks]

(c) Refer to your answer in (b), determine what value of the load resistance is the power maximum? [2 marks]

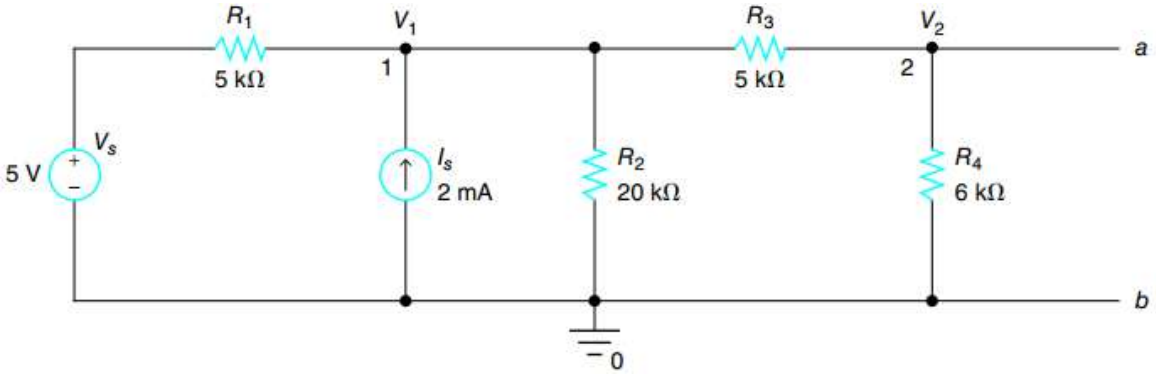


Figure 3

Question 4 [20 marks]

(CLO 2, PLO 2, C3)

(a) Referring to Figure 4, solve equivalent inductance. [4 marks]

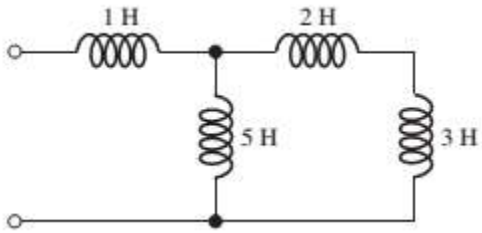


Figure 4

(b) Referring to Figure 5, using the Superposition theorem, solve the voltage V_1. [16 marks]
Please show all necessary diagrams.

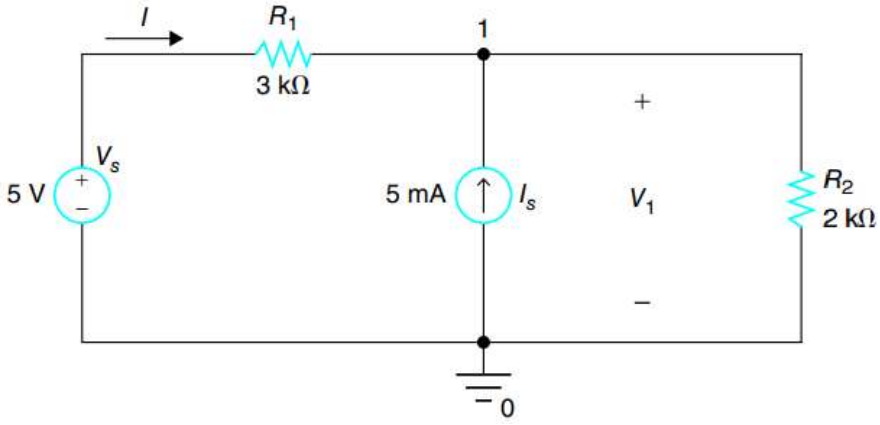


Figure 5

Question 5 [22 marks]

(CLO 2, PLO 2, C3)

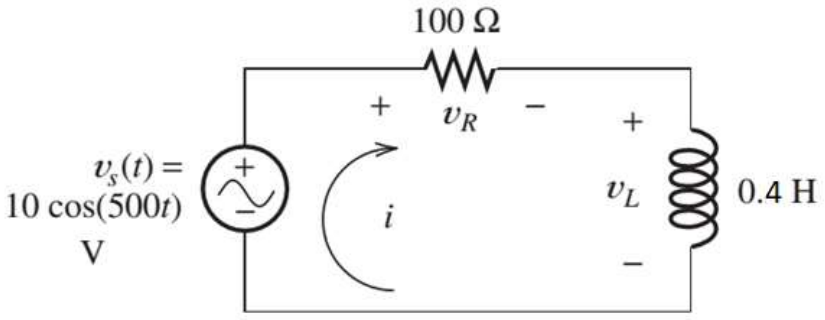


Figure 6

Referring to Figure 6,

(a) Produce the phasors for the current **I** and for the voltages **Vs, V_R, V_L** of the circuit. [12 marks]

(b) Present a phasor diagram showing **Vs, V_R, V_L** , current **I**. [8 marks]

(c) Assess the relationship between **V_R,** and current **V_L**. [2 marks]

- **END OF QUESTION -**